Model C Fit to the Real Data

Vera Schroeder

October 2026

1 Summary

Model C (hierarchical over sessions, one group per hand, distractor amplitude as a covariate) was fit to all 12,937 trials of the Sirius data (42 sessions). The main results:

- **The PSE shift grows with distractor amplitude.** The hand-level distractor coefficient is clearly positive, $\mu_{\beta_2} = 0.90$ [0.71, 1.09] (left) and 1.18 [0.99, 1.37] (right). For a typical session the bimanual PSE lies 1.2 (left) and 1.7 (right) μm below the unimanual one with a 6 μm distractor, and 6.6 and 8.9 μm below with 32 μm. Averaged over the 42 sessions the shift is -4.5 μm (left) and -5.7 μm (right), close to Model B's estimates.
- **The fit is reliable but the model misfits in specific places.** The sampler converged without problems. However, the bimanual trials at the weakest stimuli contain more "high" responses than Model C predicts, and LOO prefers Model B by 37.1 (SE 11.9), with most of the difference at the 6 and 12 μm stimuli.
- **The misfit follows from three assumptions of Model C**: unimanual and bimanual trials share their lapse rates, they share their slope, and the PSE shift is proportional to the distractor amplitude, so it shrinks to 0 for a weak distractor. For the right hand, the 6 μm sessions shift by 3.9 μm on average under Model B but only 1.7 μm under Model C.
- The amplitude dependence itself is robust, but its size at small amplitudes, and the shared lapse rates and slopes, should be revisited before Model C's estimates are used for inference (Section 6).

Every figure and number in this report is produced by `Main_Analysis_C.py`; the cell "Numbers for Fit_C_Report.tex" prints the numbers that the earlier cells do not. All intervals are 95% highest density intervals (HDIs). PSE and JND are in micrometres of stimulus amplitude: $\text{PSE}_{\mu m} = 28 + 14.63\,\text{PSE}$ and $\text{JND}_{\mu m} = 14.63\,\text{JND}$, where 28 and 14.63 are the mean and sd of the 8 amplitude levels.

2 Model and Fit

The model is the one described in the Model C convergence report. For hand j and session s,

$$\psi(x, d) = \gamma_{h,js} + (1 - \gamma_{h,js} - \gamma_{l,js})\ \text{logit}^{-1}\left(\beta_{0,js} + \beta_{1,js}\, x + \beta_{2,js}\, d\right),$$

where x is the standardized stimulus amplitude and d the distractor amplitude divided by 14.63 ($d = 0$ on unimanual trials; every session uses a single distractor amplitude on its bimanual trials).

The five session parameters are drawn from hand-level normal (for β) and logit-normal (for γ) distributions. The priors on β_0, β_1 and the lapse rates are those of Model B, and $\mu_{\beta_2} \sim \mathcal{N}(0, 1)$, $\sigma_{\beta_2} \sim \text{HalfNormal}(1)$.

Because $\beta_2 d$ only shifts the curve, the PSE shift of a session is $-\beta_2 d_s/\beta_1$ standardized units, and its JND and lapse rates are the same in both conditions.

The fit used NUTS with 4 chains of 1000 tuning and 5000 kept draws (20,000 draws in total) and took 9 minutes. The data have 3,230–3,237 trials in each hand × condition cell.

2.1 Sampling diagnostics

There were 2 divergences in the 20,000 draws, and the tree depth never exceeded 5. All 1,300 variables, including the raw sampler variables, have $\hat{R} = 1.000$, bulk ESS $\geq$ 3,410 and tail ESS $\geq$ 5,231. Unlike Model B, whose raw `z_sig_b0` variables showed $\hat{R}$ warnings from an unidentified sign, Model C defines its HalfNormal sds directly, so no such warning appears.

2.2 Posterior predictive check

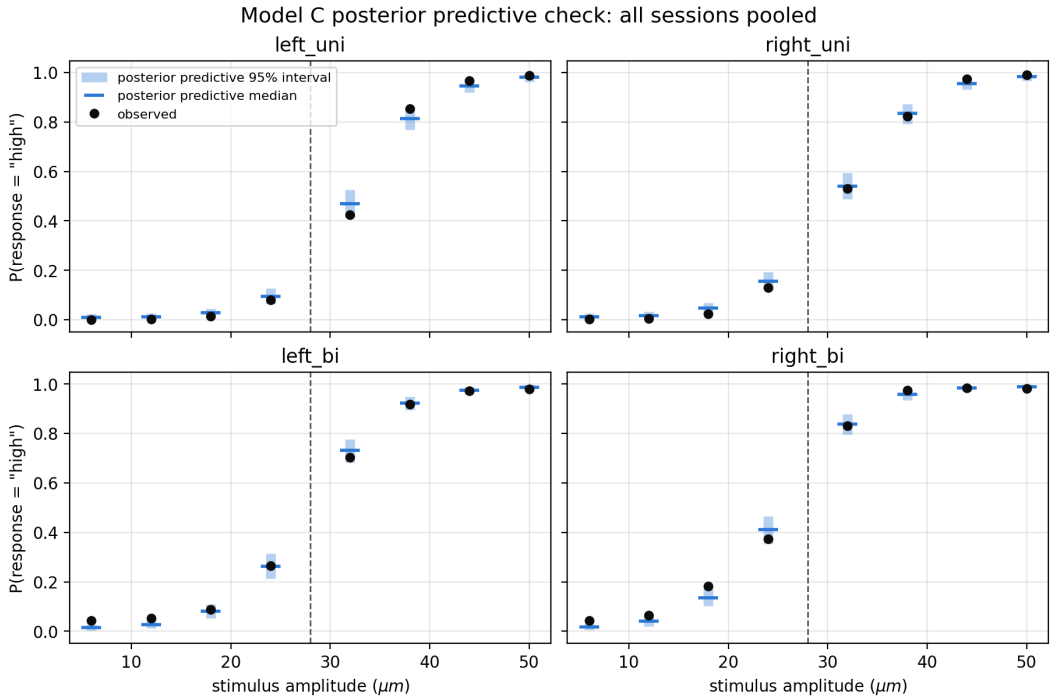


Figure 1: Observed proportion of "high" responses at each amplitude, pooled over sessions (black), against the posterior predictive median and 95% interval (blue), by hand and condition. The dashed line is the trained threshold of 28 μm.

Of the 32 hand × condition × amplitude proportions, 26 lie inside their 95% posterior predictive intervals (Figure 1). The six outside are all at the weak end of the stimulus range (Table 1): on bimanual trials Model C predicts too few "high" responses at 6–18 μm, and on left unimanual trials at 6 μm it predicts a few where the monkey gave none.

Table 1: Proportions outside their 95% posterior predictive interval.

hand	condition	amplitude (μm)	observed	predicted median	95% interval
left	unimanual	6	0.000	0.010	[0.002, 0.025]
left	bimanual	6	0.042	0.015	[0.002, 0.027]
left	bimanual	12	0.052	0.028	[0.012, 0.045]
right	bimanual	6	0.042	0.017	[0.005, 0.032]
right	bimanual	12	0.065	0.040	[0.020, 0.062]
right	bimanual	18	0.183	0.136	[0.101, 0.175]

This is the pattern expected if the distractor raises the lapse rate γ_h but the model can only use one lapse rate for both conditions. The fitted typical lapse rates ($\gamma_h = 0.009$ and 0.011, Table 3) lie between the near-zero unimanual and the higher bimanual rates that Model B estimates separately: too high for unimanual trials and too low for bimanual ones.

3 Population-level Results

Table 2: Hand-level hyperparameters: posterior mean [95% HDI]. The lapse-rate hyperparameters are on the logit scale.

	left	right
μ_{β_0}	-1.30 [-1.56, -1.04]	-0.92 [-1.29, -0.55]
μ_{β_1}	4.39 [3.95, 4.85]	4.25 [3.92, 4.62]
μ_{β_2}	0.90 [0.71, 1.09]	1.18 [0.99, 1.37]
σ_{β_0}	0.60 [0.39, 0.81]	1.07 [0.79, 1.36]
σ_{β_1}	0.90 [0.57, 1.26]	0.35 [0.00, 0.69]
σ_{β_2}	0.30 [0.08, 0.51]	0.23 [0.00, 0.45]
μ_{γ_h}	-3.33 [-4.07, -2.69]	-3.11 [-3.91, -2.43]
μ_{γ_l}	-3.42 [-4.36, -2.60]	-3.41 [-4.18, -2.67]
σ_{γ_h}	0.34 [0.00, 0.82]	0.34 [0.00, 0.82]
σ_{γ_l}	0.40 [0.00, 0.96]	0.47 [0.00, 1.03]

Table 3: A typical session (all parameters at their hand-level means): unimanual PSE, JND and lapse rates, posterior mean [95% HDI]. The JND and lapse rates are the same on bimanual trials in Model C.

	left	right
PSE, unimanual (μm)	32.34 [31.51, 33.17]	31.15 [29.92, 32.40]
JND (μm)	3.75 [3.39, 4.11]	3.88 [3.59, 4.19]
γ_h	0.0091 [0.0033, 0.0147]	0.0113 [0.0042, 0.0189]
γ_l	0.0086 [0.0023, 0.0155]	0.0085 [0.0029, 0.0143]

The unimanual PSE of a typical session lies 4.3 (left) and 3.2 (right) μm above the trained threshold of 28 μm (Table 3). The sessions differ in their bias ($\sigma_{\beta_0} = 0.60$ and 1.07) and, for the left hand,

in their slope ($\sigma_{\beta_1} = 0.90$): the left-hand session JNDs range from 2.95 to 6.59 μm, the right-hand ones only from 3.65 to 4.31 μm. As in Model B and in Model C's convergence study, σ_{γ_h} and σ_{γ_l} are not identified; their posteriors largely reflect the prior.

The data are informative about μ_{β_2}: its posterior sd is 0.097 for both hands, against a prior sd of 1.

4 Effect of the Distractor

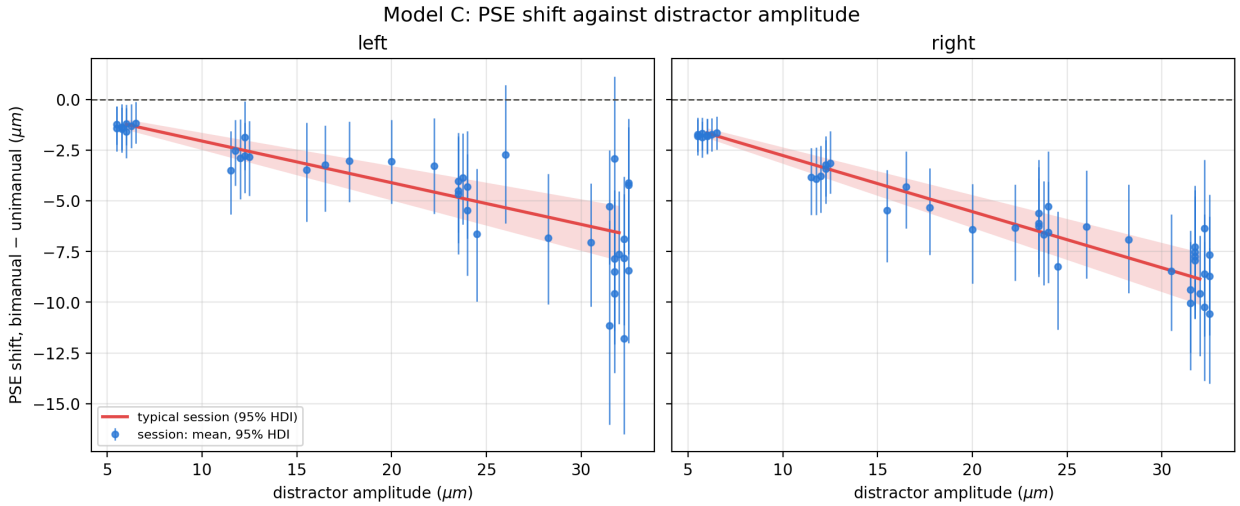


Figure 2: PSE shift (bimanual − unimanual) against the session's distractor amplitude. Blue: every session, posterior mean and 95% HDI (sessions sharing an amplitude are spread slightly). Red: a typical session, posterior mean and 95% HDI.

Table 4: PSE shift (bimanual − unimanual, μm) of a typical session at the most used distractor amplitudes: posterior mean [95% HDI].

distractor (μm)	left	right
6	−1.23 [−1.49, −0.98]	−1.66 [−1.90, −1.41]
12	−2.46 [−2.98, −1.97]	−3.32 [−3.79, −2.83]
24	−4.93 [−5.96, −3.93]	−6.64 [−7.59, −5.65]
32	−6.57 [−7.95, −5.24]	−8.85 [−10.12, −7.54]

The distractor lowers the PSE, and by more the stronger it is (Figure 2, Table 4). The shift is larger for the right hand at every amplitude. Averaged over the 42 sessions it is −4.51 [−5.15, −3.86] μm for the left hand and −5.74 [−6.39, −5.11] μm for the right. At the session level, the shift's HDI lies below 0 in 40 of 42 sessions (left) and all 42 (right).

The sessions vary only a little around the hand-level line: σ_{β_2} is 0.30 (left) and 0.23 (right), so the spread of the session shifts in Figure 2 mostly reflects the sessions' different distractor amplitudes rather than different β_2. The convergence study showed that a session's β_2 is poorly determined when its distractor is weak and is then pulled toward the hand mean; the 6 μm sessions in Figure 2 sit almost exactly on the line.

By construction, Model C cannot show an effect of the distractor on the JND or the lapse rates.

5 Comparison with Model B

Model B fits unimanual and bimanual trials as separate groups, each with its own per-session parameters, and has no amplitude covariate. Both models were fit to the same trials in the same order.

Leave-one-out cross-validation. Model C has $\mathrm{elpd}_{\mathrm{loo}} = -3217.0$ (SE 65.7) with $p_{\mathrm{loo}} = 138.6$ and every Pareto k at most 0.33; Model B has -3180.0 (SE 64.3) with $p_{\mathrm{loo}} = 178.7$. Model C is worse by 37.1 (SE 11.9), 3.1 standard errors, despite needing fewer effective parameters.

Table 5: $\mathrm{elpd}_{\mathrm{loo}}(\mathrm{C}) - \mathrm{elpd}_{\mathrm{loo}}(\mathrm{B})$ summed over the trials at each amplitude, by hand and condition. Negative values favor Model B.

	left		right	
amplitude (μm)	bimanual	unimanual	bimanual	unimanual
6	−6.6	−3.5	−5.1	−2.3
12	−4.2	−3.0	−1.8	−2.8
18	0.7	−1.6	−3.4	−0.3
24	−1.0	0.8	1.9	0.8
32	−0.1	−3.4	0.3	1.7
38	−0.3	−1.4	0.2	1.6
44	0.0	−3.8	0.4	−1.5
50	−0.1	0.3	0.1	0.0

Model C loses 29.2 of its 37.1 at the 6 and 12 μm stimuli (Table 5): 17.6 on bimanual and 11.6 on unimanual trials. This is the shared-lapse misfit of Section 2.2, seen from both sides. The next largest losses are on left unimanual trials at 32 and 44 μm (-3.4 and -3.8), on the rising part of the curve. With one slope for both conditions, Model C cannot give the left-hand unimanual curve a different steepness than the bimanual one, which Model B allows.

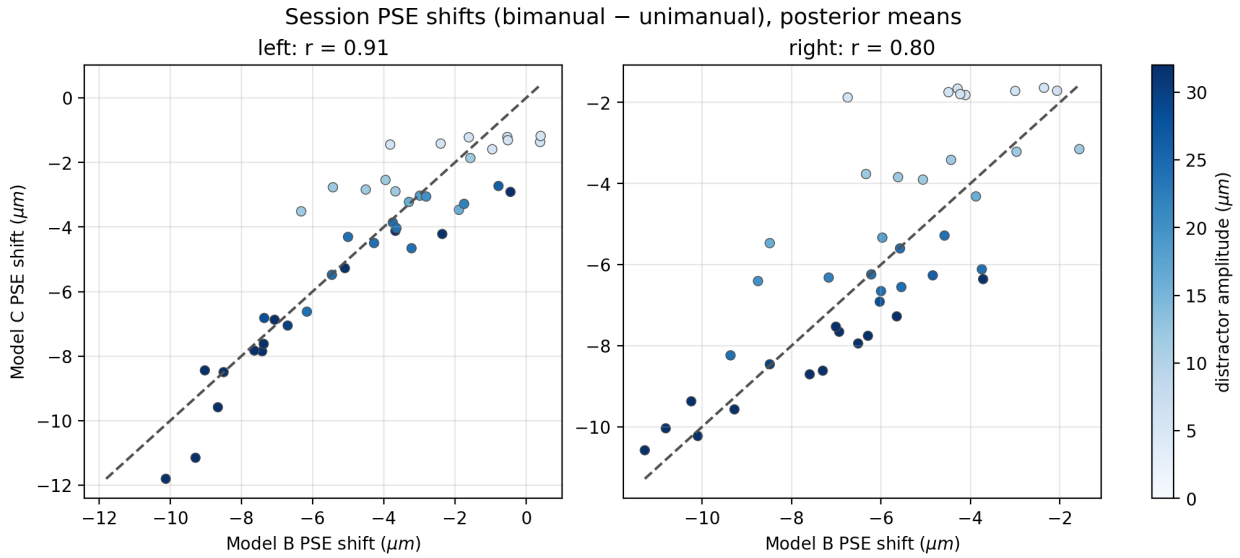


Figure 3: Session PSE shifts (bimanual − unimanual), posterior means: Model C against Model B, colored by the session's distractor amplitude. The dashed line is the identity; r is the correlation.

Session PSE shifts. The two models agree on the average shift (-4.51 against -4.29 μm for the left hand, -5.74 against -6.06 for the right) and, for the left hand, closely on the session shifts ($r = 0.91$, Figure 3). For the right hand ($r = 0.80$) they disagree at the weakest distractor:

Table 6: Session PSE shifts (μm) at the weakest and strongest distractor: mean and range of the posterior means over the sessions with that amplitude.

hand	distractor	Model B	Model C
left	6 μm (8 sessions)	-1.13 (-3.82 to 0.41)	-1.34 (-1.59 to -1.18)
left	32 μm (13 sessions)	-6.67 (-10.12 to -0.44)	-7.40 (-11.80 to -2.91)
right	6 μm (8 sessions)	-3.91 (-6.74 to -2.06)	-1.75 (-1.88 to -1.64)
right	32 μm (13 sessions)	-7.90 (-11.28 to -3.72)	-8.58 (-10.58 to -6.36)

For the right hand, Model B estimates the 6 μm sessions to shift by 3.91 μm on average, Model C by only 1.75 (Table 6). Model C's shift is proportional to the amplitude, so a weak distractor must give a small shift, and with β_2 poorly determined in these sessions they all fall onto the hand-level line. Model B, which estimates each session's shift freely, suggests that for the right hand even a weak distractor shifts the PSE substantially: an effect of the distractor being present, on top of its amplitude.

6 Conclusions

- The fit is reliable: 2 divergences in 20,000 draws, $\hat{R} = 1.000$ and bulk ESS $\geq$ 3,410 for every variable, every Pareto k good.
- The PSE shift grows with distractor amplitude for both hands ($\mu_{\beta_2} > 0$ with posterior probability 1.000), more steeply for the right hand. This is the main new result of Model C, and it supports the exploratory amplitude trend in the Model B report.

- Model C predicts the data worse than Model B (elpd difference −37.1, SE 11.9). The misfit sits where three of its assumptions meet the data:
    1. one lapse rate for both conditions, while the distractor raises γ_h: too few "high" responses are predicted at weak stimuli on bimanual trials, and too many on unimanual ones;
    2. one slope for both conditions, which costs fit on the rising part of the left-hand unimanual curve;
    3. a shift proportional to amplitude: for the right hand at 6 μm, Model C's shift is less than half of Model B's.
- The size of the shift at weak distractors, and Model C's lapse rates and JNDs, should therefore not be taken at face value until these assumptions are relaxed.

Recommended revisions.

1. Lapse rates per hand and condition (unimanual, bimanual). The evidence for this is the strongest.
2. A distractor-presence term, $\beta_3 \cdot \mathbb{1}[\text{bimanual}]$, so that the PSE shift is $-(\beta_3 + \beta_2 d)/\beta_1$ rather than proportional to d.
3. Possibly a separate slope on bimanual trials, which would let the distractor change the JND.

Each revision changes the model, so it would need a new convergence study before being fit to the real data again.